

1. (Fall 2024, #9) Evaluate the following integrals.

$$(a) \int_1^e \frac{\ln x}{x\sqrt{1+(\ln x)^2}} dx$$

$$u = 1 + (\ln x)^2 \Rightarrow du = 2(\ln x) \cdot \frac{1}{x} dx \Rightarrow dx = \frac{x}{2 \ln x} \cdot du$$

$$u(1) = 1 + (\ln(1))^2 = 1$$

$$u(e) = 1 + (\ln(e))^2 = 2$$

$$\int_1^e \frac{\ln x}{x\sqrt{1+(\ln x)^2}} dx = \int_1^2 \frac{\cancel{\ln x}}{\cancel{x} \cdot \sqrt{u}} \cdot \frac{\cancel{x}}{2 \cancel{\ln x}} du = \int_1^2 \frac{1}{2\sqrt{u}} du = \sqrt{u} \Big|_1^2 = \sqrt{2} - 1$$

$$(b) \int e^{xe^x} (e^x + xe^x) dx$$

$$u = xe^x \Rightarrow du = (e^x + xe^x) dx \Rightarrow dx = \frac{1}{e^x + xe^x} du$$

$$\int e^{xe^x} (e^x + xe^x) dx = \int e^u \cancel{(e^x + xe^x)} \cdot \frac{1}{\cancel{e^x + xe^x}} du = \int e^u du = e^u + C = e^{xe^x} + C$$

2. (Spring 2024, #8) Evaluate the following integrals.

(a)

$$\int \tan(x) \ln(\cos(x)) dx$$

(Hint: what is the simplified derivative of $\ln(\cos(x))$?)

$$u = \ln(\cos(x)) \Rightarrow du = \frac{1}{\cos(x)} \cdot (-\sin(x)) dx = -\tan(x) dx \Rightarrow dx = \frac{-1}{\tan(x)} du$$

$$\int \tan(x) \ln(\cos(x)) dx = \int \cancel{\tan(x)} \cdot u \cdot \frac{-1}{\cancel{\tan(x)}} du = -\int u du = -\frac{u^2}{2} + C$$

$$= -\frac{(\ln(\cos(x)))^2}{2} + C$$

(b)

$$\int_0^1 \frac{2x + e^{3x}}{3x^2 + e^{3x} + 3} dx$$

$$u = 3x^2 + e^{3x} + 3 \Rightarrow du = (6x + 3e^{3x}) dx \Rightarrow dx = \frac{1}{3(2x + e^{3x})} du$$

$$u(0) = 3(0)^2 + e^{3(0)} + 3 = 4$$

$$u(1) = 3(1)^2 + e^{3(1)} + 3 = 6 + e^3$$

$$\int_0^1 \frac{2x + e^{3x}}{3x^2 + e^{3x} + 3} dx = \int_4^{e^3+6} \frac{\cancel{2x + e^{3x}}}{u} \cdot \frac{1}{3(\cancel{2x + e^{3x}})} du = \frac{1}{3} \int_4^{e^3+6} \frac{1}{u} du$$

$$= \frac{1}{3} \ln|u| \Big|_4^{e^3+6} = \frac{1}{3} (\ln(e^3+6) - \ln(4))$$

3. (Fall 2023, #8) Compute the following indefinite/definite integrals.

(a) $\int \frac{x^{3/2} - 5x + 2}{x^{1/2}} dx$

No substitution is needed!

$$\int \frac{x^{3/2} - 5x + 2}{x^{1/2}} dx = \int (x - 5x^{1/2} + 2x^{-1/2}) dx = \frac{x^2}{2} - \frac{10}{3} x^{3/2} + 4x^{1/2} + C$$

(b) $\int \frac{x}{\sqrt{x-2}} dx$

$$u = x - 2 \Rightarrow du = dx$$

$$\begin{aligned} \int \frac{x}{\sqrt{x-2}} dx &= \int \frac{u+2}{\sqrt{u}} du = \int (u^{1/2} + 2u^{-1/2}) du = \frac{2}{3} u^{3/2} + 4u^{1/2} + C \\ &= \frac{2}{3} (x-2)^{3/2} + 4(x-2)^{1/2} + C \end{aligned}$$

(c) $\int_0^{\ln 5} \frac{e^x dx}{2e^x + 5}$

$$u = 2e^x + 5 \Rightarrow du = 2e^x dx \Rightarrow dx = \frac{1}{2e^x} \cdot du$$

$$u(0) = 2e^0 + 5 = 7$$

$$u(\ln 5) = 2e^{\ln 5} + 5 = 15$$

$$\int_0^{\ln 5} \frac{e^x}{2e^x + 5} dx = \int_7^{15} \frac{e^x}{u} \cdot \frac{1}{2e^x} \cdot du = \frac{1}{2} \int_7^{15} \frac{1}{u} du = \frac{1}{2} \ln|u| \Big|_7^{15} = \frac{1}{2} (\ln 15 - \ln 7) = \frac{1}{2} \ln\left(\frac{15}{7}\right)$$

4. (Spring 2023, #8) Evaluate the following integrals.

(a) $\int \frac{\cot(x)}{\ln(\sin(x))} dx$

$$u = \ln(\sin(x)) \Rightarrow du = \frac{1}{\sin(x)} \cos(x) \cdot dx \Rightarrow du = \cot(x) \cdot dx \Rightarrow dx = \frac{1}{\cot(x)} du$$

$$\int \frac{\cot(x)}{\ln(\sin(x))} dx = \int \frac{\cancel{\cot(x)}}{u} \cdot \frac{1}{\cancel{\cot(x)}} du = \int \frac{1}{u} du = \ln|u| + C = \ln|\ln(\sin(x))| + C$$

(b) $\int_0^{\ln(\pi)} (e^{3e^{3x}+3x}) dx = \int_0^{\ln(\pi)} (e^{3e^{3x}} \cdot e^{3x}) dx$

$$u = e^{3x} \Rightarrow du = 3e^{3x} dx \Rightarrow dx = \frac{1}{3e^{3x}} du$$

$$u(0) = e^{3(0)} = 1$$

$$u(\ln(\pi)) = e^{3\ln(\pi)} = \pi^3$$

$$\int_0^{\ln(\pi)} (e^{3e^{3x}} \cdot e^{3x}) dx = \int_1^{\pi^3} e^{3u} \cdot \cancel{e^{3x}} \cdot \frac{1}{3\cancel{e^{3x}}} du = \int_1^{\pi^3} \frac{e^{3u}}{3} du = \frac{1}{9} e^{3u} \Big|_1^{\pi^3}$$

$$= \frac{1}{9} (e^{3\pi^3} - e^3)$$

5. (Fall 2022, #7) Find the following explicitly:

$$\begin{aligned}
 \text{(a)} \quad \int_0^4 |x-2| dx &= \int_0^2 (2-x) dx + \int_2^4 (x-2) dx \\
 &= \left[2x - \frac{x^2}{2} \right]_0^2 + \left[\frac{x^2}{2} - 2x \right]_2^4 \\
 &= (2 - 0) + (0 - (-2)) = 4
 \end{aligned}$$

$$\text{(b)} \quad \int_0^{1/2} x \sec^2(\pi x^2) dx$$

$$u = \pi x^2 \Rightarrow du = 2\pi x \cdot dx \Rightarrow dx = \frac{1}{2\pi x} du$$

$$u(0) = \pi(0)^2 = 0$$

$$u(1/2) = \pi(1/2)^2 = \frac{\pi}{4}$$

$$\begin{aligned}
 \int_0^{1/2} x \sec^2(\pi x^2) dx &= \int_0^{\pi/4} x \cdot \sec^2(u) \cdot \frac{1}{2\pi x} \cdot du = \frac{1}{2\pi} \int_0^{\pi/4} \sec^2(u) du = \frac{1}{2\pi} [\tan(u)]_0^{\pi/4} \\
 &= \frac{1}{2\pi} \left(\tan\left(\frac{\pi}{4}\right) - \tan(0) \right) = \frac{1}{2\pi} (1 - 0) = \frac{1}{2\pi}
 \end{aligned}$$

$$\text{(c)} \quad \int \frac{1}{x \ln(x)} dx$$

$$u = \ln(x) \Rightarrow du = \frac{1}{x} dx \Rightarrow dx = x \cdot du$$

$$\int \frac{1}{x \ln(x)} dx = \int \frac{1}{x \cdot u} \cdot x \cdot du = \int \frac{1}{u} du = \ln|u| + C = \ln|\ln x| + C$$